

PLANETVORTEX

PLANETVORTEX — Theorem C — the rigid heptagon lattice

The theorem edition of the planetary gravity-geometry bench — mini-repository planetvortex/ of the TRIVORTEX framework, version 1.0.0. This dossier binds one statement to its registered protocol; the full science lives in the big monograph (docs/monograph/).

Statement

The seven-vortex regular heptagon ($\Gamma_i = \Gamma$, circumradius R) is a relative equilibrium rotating rigidly at $\omega_7 = \frac{\Gamma(N-1)}{4\pi R^2} = \frac{3\Gamma}{2\pi R^2} = 0.4774648292756860\dots$; its co-rotating spectrum is Havelock-stable with $\max Re \lambda = 7.0 \times 10^{-9}$ on the numerically-zero floor (classifier 10^{-7}) — the last stable level before the threshold at $N=8$; and its Hamiltonian is exactly $H_{ring} = -\frac{7\Gamma^2}{2\pi} \ln(\sqrt{7} R^3) = -1.08395426662194\dots$ at $\Gamma = R = 1$.

Proof

The rotation formula is the roots-of-unity register at $N=7$: each vortex feels the six others through the chord sum, which evaluates to $(N-1)\Gamma/(4\pi R^2)$. The stability classification is the Havelock register: the co-rotating eigenvalues of the N -gon are purely imaginary for $N \leq 7$ and cross into the right half-plane at $N=8$ (+0.4502 in the sibling's scan). The Hamiltonian: the 21 pairwise distances are seven of each chord class, so $\sum_{i < j} \ln d_{ij} = 7 \ln(R^3 \sqrt{7})$ by Theorem A, giving $H_{ring} = 7 H_{cell}$. ◻

Protocol binding

item	value
stage	P5
recorded registers	$1.35 \times 10^{-12} / 7.0 \times 10^{-9}$
protocol file	results/protocols/ (default preset)
test guard	tests/ re-derives every preset-independent register

Cross-references

- the parent framework: Theorem 3.1 and the three-vortex anchor;
- the sibling bench polyvortex: the W2/W4 registers at $N=7$ — the same heptagon, cross-pinned bit-for-bit;
- the big monograph: Sections 3 (Theorem A), 6 (Theorem C), 7 (Theorem B), 4 (Lemma D), 8 (Lemma E);
- the formalization queue: roadmap item V4 (Coq / Lean 4).